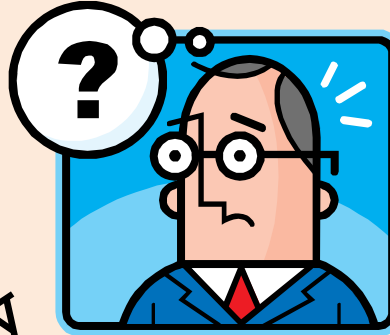


Information Quality Principles and Domains

John R. Talburt, PhD, IQCP

What is Information Quality?

Accurate Data?



**Complete and Timely
Information?**

**Information
free of defects?**

**Application of TQM
to information?**

***RIGHT
INFORMATION,
RIGHT PLACE, AT
THE RIGHT TIME?***

MANY DEFINITIONS

Information Quality is About

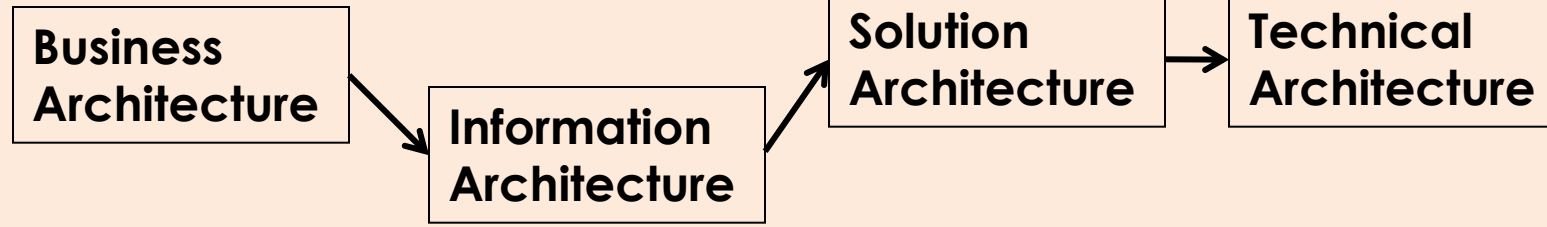
- Maximizing the value and minimizing the risk of an organization's information assets
- Assuring the information products produced by the organization create value for the people and systems who use them



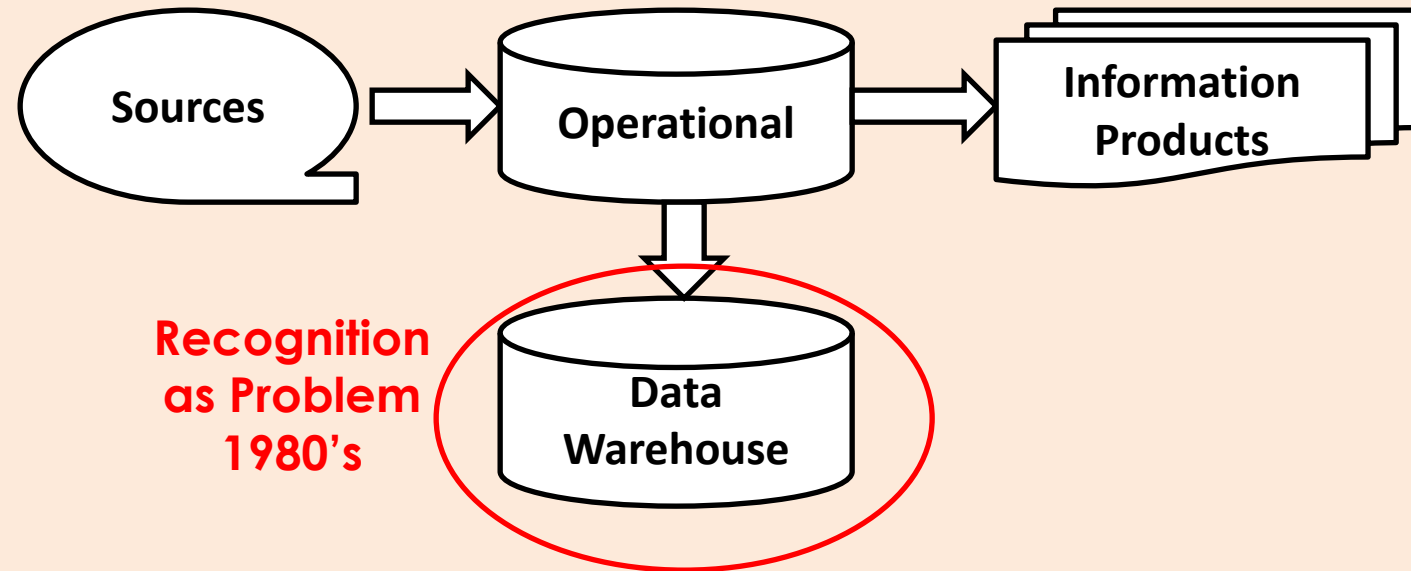
HISTORY OF INFORMATION QUALITY

Problem Recognition: The Data Cleansing Phase

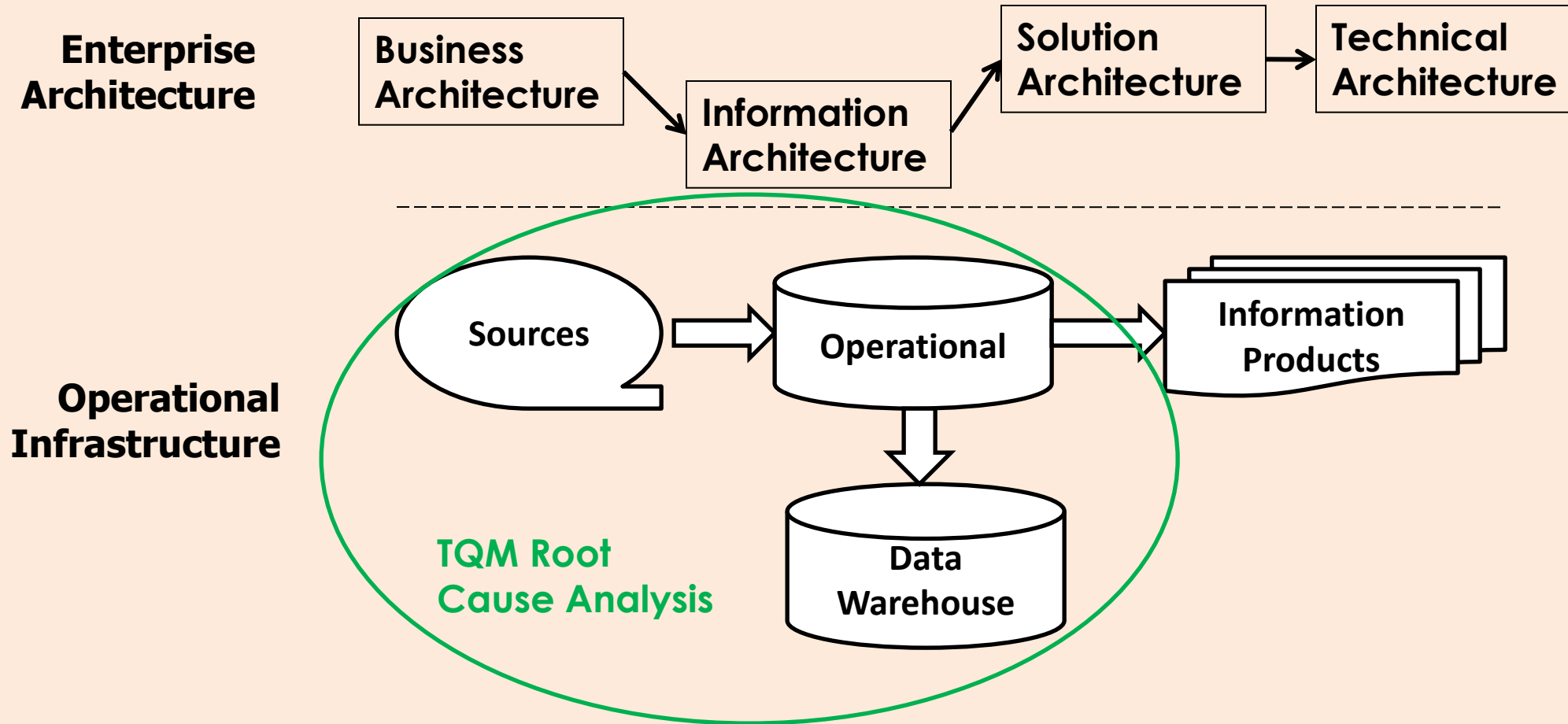
**Enterprise
Architecture**



**Operational
Infrastructure**

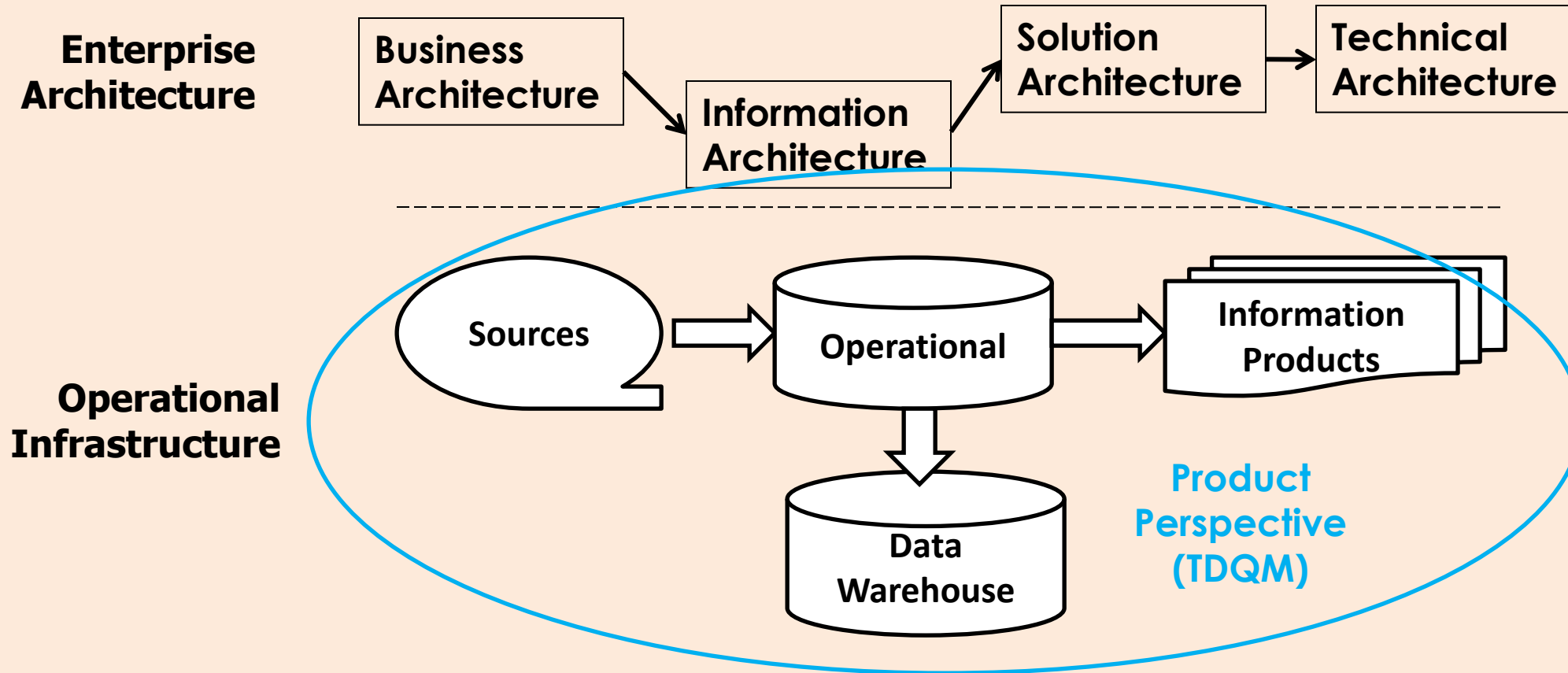


Root Cause Detection: The Prevention Phase

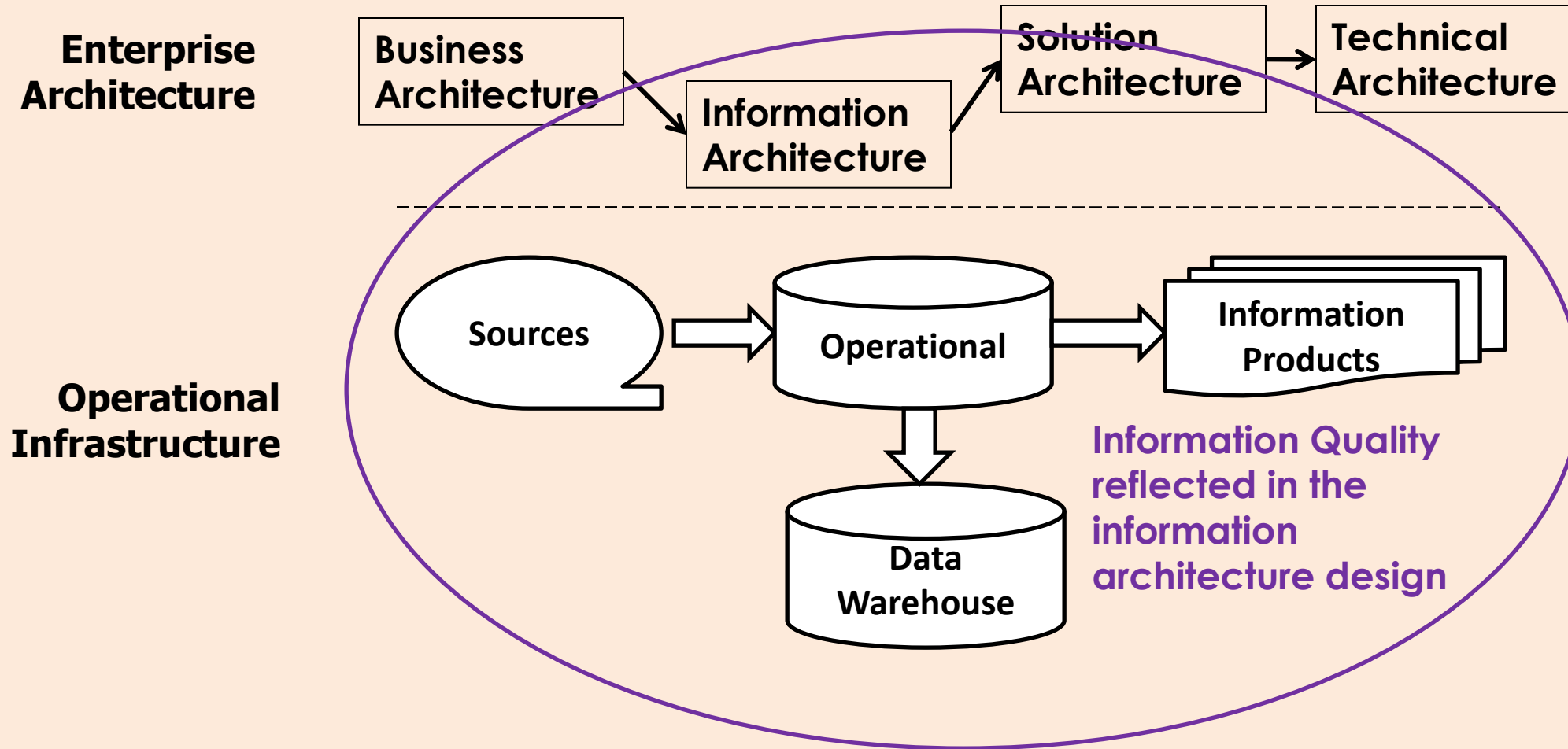


Manufacturing Analogy:

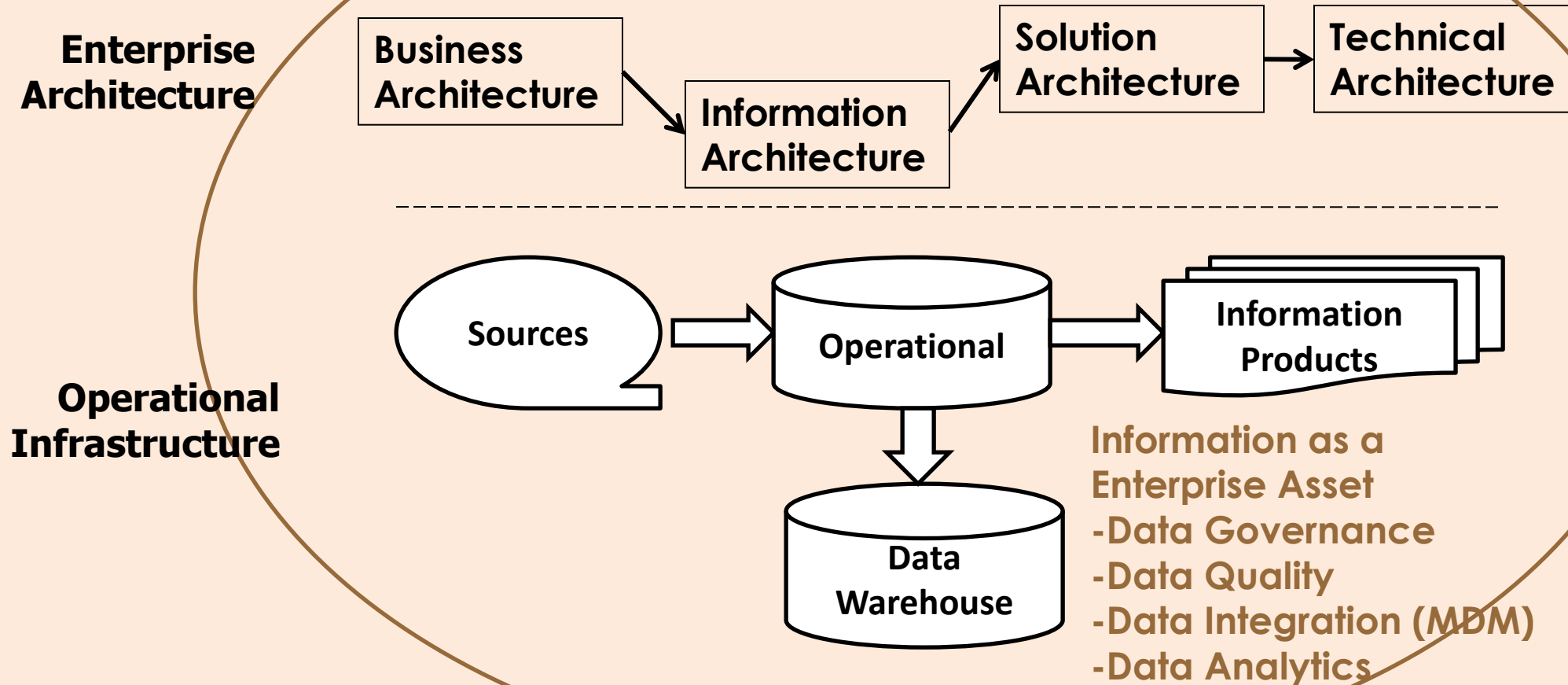
Information Product & Process Management Phase



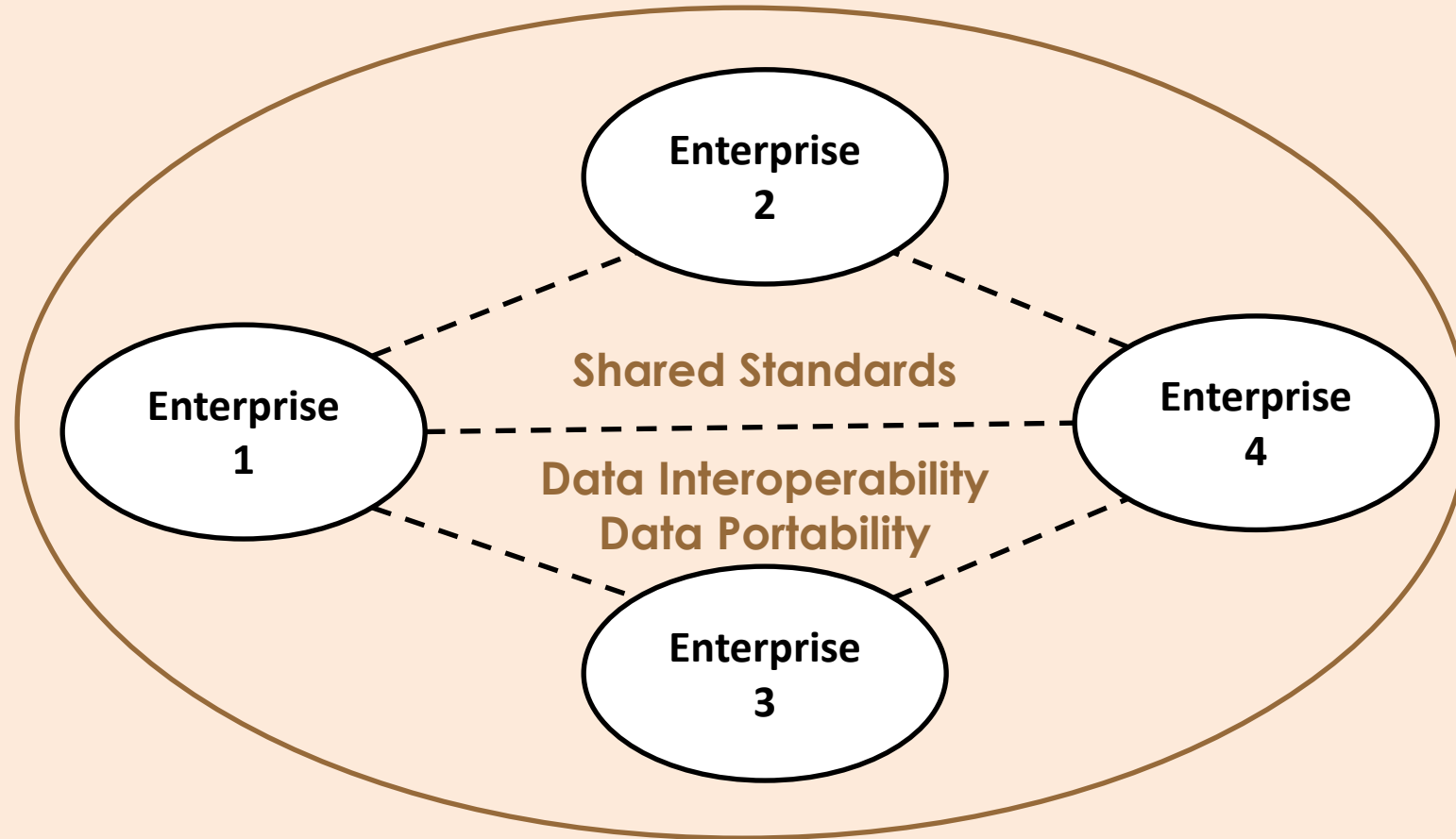
Information Architecture View: The Quality by Design Phase



Enterprise View: Information as an Organizational Asset



What's Next?



Emerging Topics in IQ

- ISO 8000 Data Quality Standards
- DQ Management for Streaming/R-T Data
- DQ for Unstructured Data
- DQ for Internet of Things (IOT)
- Impact and assessment of DQ for Machine Learning training data
 - Bias, Data Poisoning, Model integrity
- Reliability and Safety of AI Systems
- Automation of DQ Assessment and Cleansing (Data Washing Machines)
- Automation of Data Governance
- Data Literacy
- Data Democratization
- Data Monetization, Infonomics

SEVEN FUNDAMENTAL PRINCIPLES OF INFORMATION QUALITY

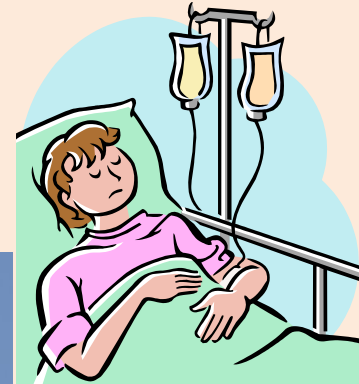
Information Quality Principle #1

Data is a resource from which individuals and organizations can **produce information products** for the purpose of **creating value**

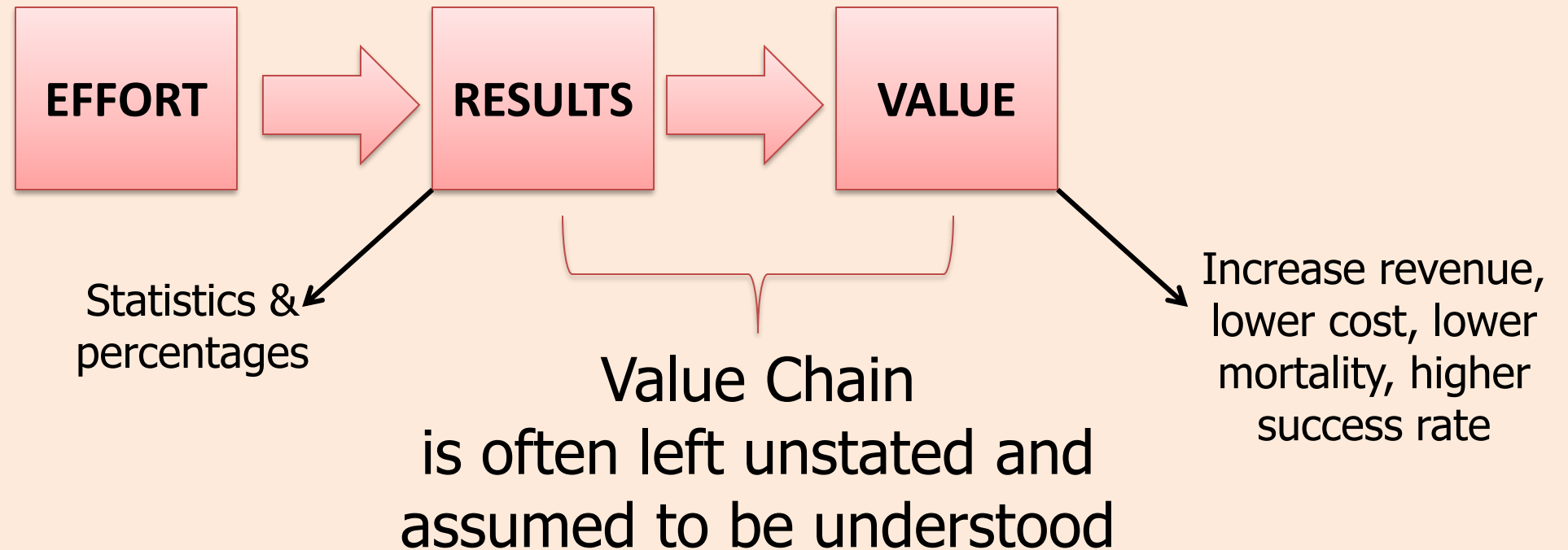
- Data collected, processed, and formatted to a particular use is an **Information Product (IP)**
- When an organization's information products create value, then data and information become an **organizational assets**

Value

- $\text{VALUE} = \text{BENEFITS} - \text{INVESTMENT}$
- Creating value should be the reason we do things
- Value can be
 - Monetary - measured in dollars
 - Human – quality of life
 - Mission Accomplishment



Don't Confuse Results with Value!



Business Example

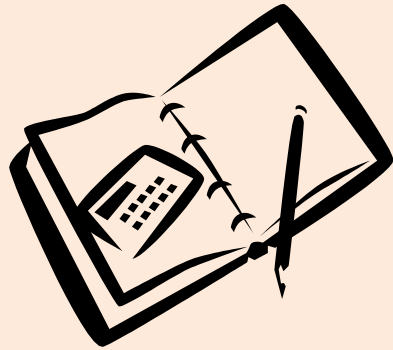
- **Effort** - 5 man-year system development with \$1 million direct cost and \$500K indirect cost
- **Result** – New customer recognition system 15% more accurate than the legacy system
- **Result** – Surveys show customer satisfaction up 20%
- **Result** – Analysis shows customer retention is up by 8%
- **Value** – Difference between customer acquisition cost and customer maintenance cost projects a net revenue increase over 5 years = \$20 million

Information Quality Principle #2

Data and information should be controlled as an organizational asset!

- Controlling data as an asset is called **Data Governance**
- Data Governance is not Data Management! - Data Governance sets the rules for making data management decisions
- Data Governance is one the fastest growing areas of information quality

The same way that every business has



An Accounting Function led by the
Chief Financial Officer (CFO)

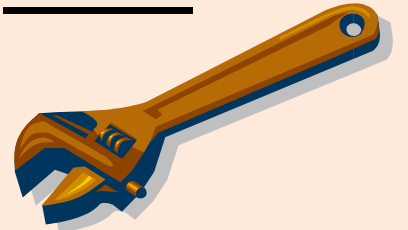
Information Technology Function led by the
Chief Information Officer (CIO)



It should also have
Information Quality and Data Governance Programs
led by the
Chief Data Officer (CDO)

Information Quality Principle #3

Information can only produce value when it is used for some purpose!



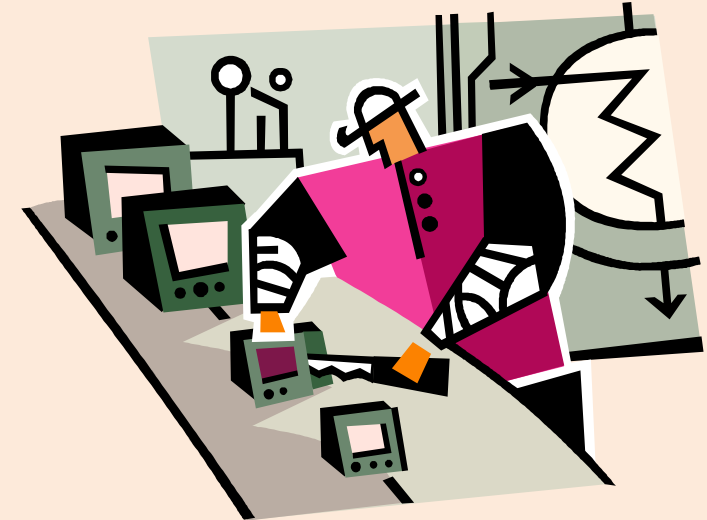
- Data transformed & organized into a form for use in an application is an **Information Product (IP)**
- The quality of an IP is measured in terms of the value it creates for the user



Information Quality Principle #4

Information should be managed as product: the product of an information system, not a by-product!

- Managing information as a system by-product will not work
 - Focus is on the systems, software and hardware
 - The means becomes the end



Successful Product Management

1. Understand information consumer needs
2. Manage information as the product of a well-defined production process
3. Manage the life cycle of the information product
4. Appoint an information product manager



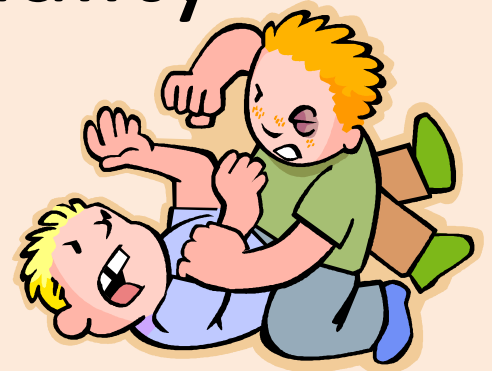
Information Quality Principles #5

Information Quality and Data Quality are related but are different in their focus

- Information Quality
 - applies to information products, including their governance and use
 - measured in value units: money, quality-of-life, mission achievement
- Data Quality
 - applies to operational data internal to the organization
 - measured in percentage: degree to which data requirements are met

Information Quality versus Data Quality

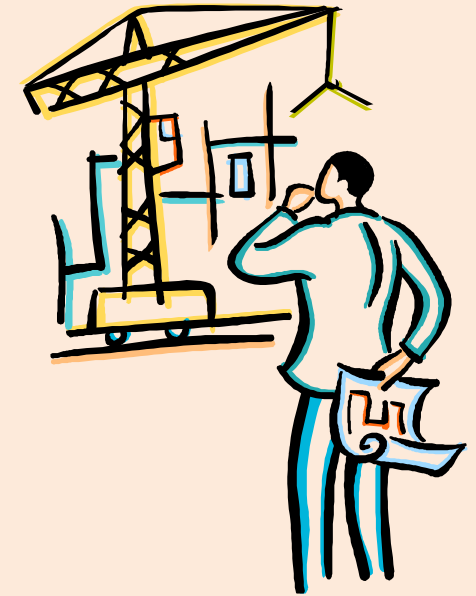
- Data Quality has an inward focus
 - About the “condition” of data
 - Data Quality is measured in conformance to data requirements
 - Data Quality measures are counts, percentages, proportions, ratios, ...
- Information Quality has an outward focus
 - About creating “value” from information
 - About meeting user requirements/expectations
 - Information Quality measures are monetary, treatment outcomes, ...



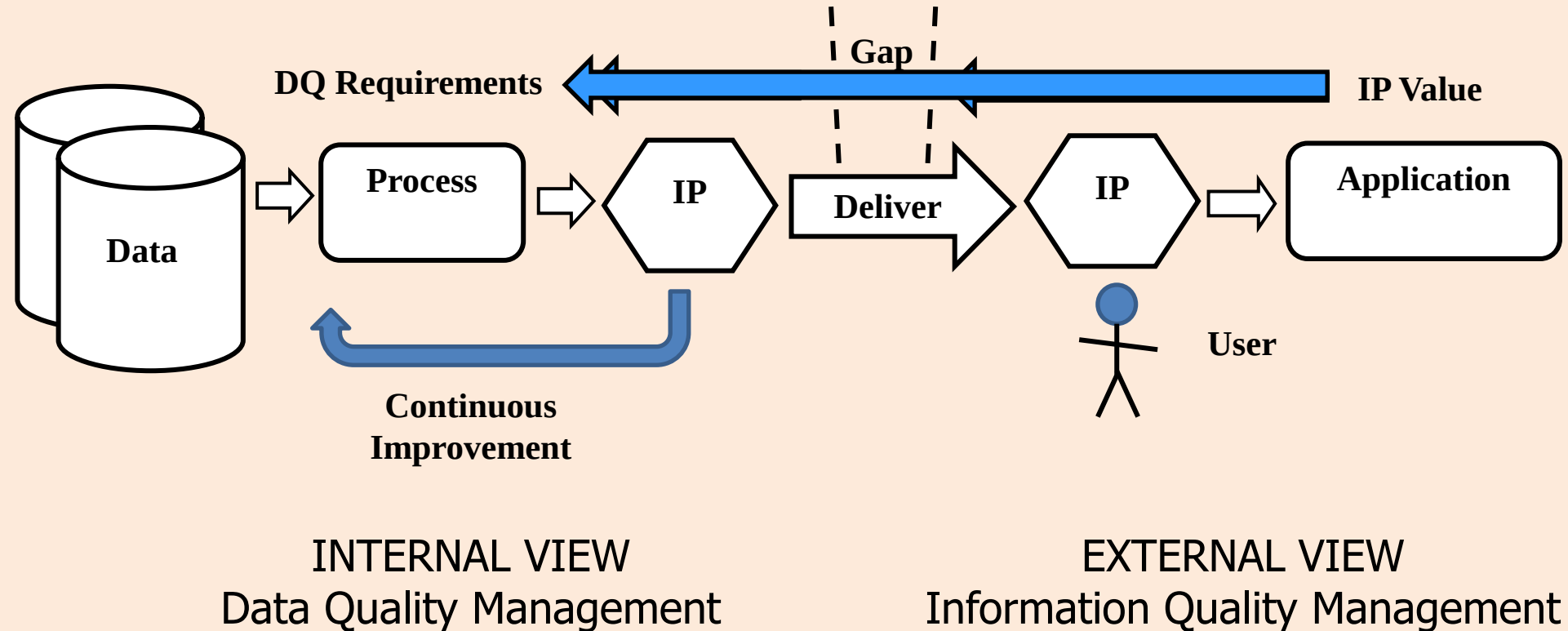
Like the Engineering Dilemma

Are you building the thing right? or Are you building the right thing?

- Data quality is assuring data conform to specification (building it right)
- Information quality is building an Information Product (IP) that creates value for the user (building the right thing)
- Our goal is to do both!!



The Data Quality – Information Quality Gap

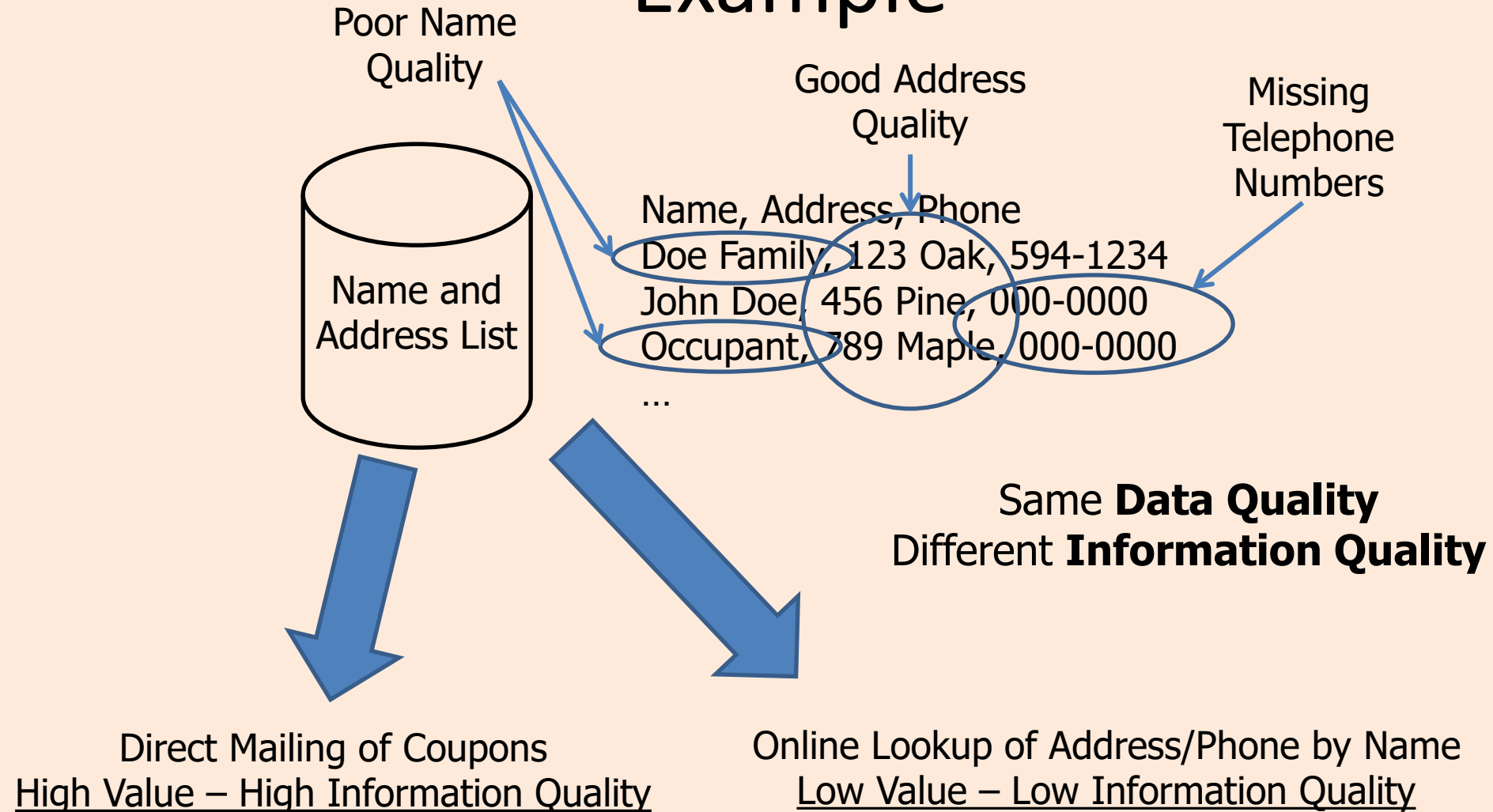


Information Quality Principle #6

Information Quality is contextual (situational), not absolute

- The same data can have different information quality when used in different information products for different purposes
- The same data with the same set of data requirements has a fixed data quality
 - DQ changes when either data or the data requirements change
 - Changes in the use of information products can change the data requirements

Example



Information Quality Principle #7

Data have many intrinsic characteristics or dimensions that can be used to express data requirements

- E.g Wang-Strong Framework of Dimensions

Intrinsic	Contextual	Representational	Accessibility
-Accuracy	-Value-added	-Interpretability	-Access
-Believability	-Relevancy	-Ease of Understanding	-Security
-Objectivity	-Timeliness	-Representational consistency	
-Reputation	-Completeness	-Conciseness of representation	
	-Amount of data	-Manipulability	

SIX DOMAINS OF INFORMATION QUALITY KNOWLEDGE AND PRACTICE

Six Domains IQ Knowledge & Practice*

Information Quality knowledge and practice comprises

- Information quality value & business impact
- Information quality strategy & governance
- Information quality environment & culture
- Information quality measurement & improvement
- Sustaining information quality
- Information architecture quality

**Data
Quality**



**IAIDQ Job Analysis Report (www.iaidq.org)*

Some Roles of IQ Graduates

- DQ/IQ Analyst, DQ/IQ Manager
- Data Governance (DG) Steward, DG Analyst/Director
- Solution Developer
- Data Developer, Data Engineer
- Data Scientist, Machine Learning Engineer
- Business Analyst
- Data Analyst, Risk Data Analyst
- Chief Data Officer

Questions and Discussion